Certified Records: An Exact Audit of AI-Claimed Mathematical Constructions

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Abstract

AI discovery systems now routinely announce "new best-known" constructions for open problems in combinatorics, geometry and analysis. The constructions are scored by floating-point evaluators, some with explicit tolerances. We collect the published artifacts of 115 record claims from ten systems and papers — AlphaEvolve, its open-source successors, test-time-training methods and a public agent leaderboard — and re-verify every one in exact rational and interval arithmetic. The constructions are almost always sound: none of AlphaEvolve's 56 released constructions is invalid, and four violate constraints by 10^{-17}–10^{-15}, which certified repairs absorb. The claims around them are weaker. Eight printed values do not hold for the released construction, most of them bounds rounded in the unsafe direction. Seven claimed records do not exceed a construction that was already public, in one case for 23 days. Two records exist only under an evaluator's tolerance, and the top two entries of a public leaderboard exploit a verifier edge case (true score 0.214, not the displayed 0.563). We also build inputs that published evaluators accept with scores above every genuine record, exceeding the true optimum by up to 1.3×10^{-5}. As a by-product we obtain certified values for all audited constructions and computer-assisted existence proofs (Krawczyk interval-Newton) for 61 efficient spherical t-designs: all twelve AlphaEvolve designs, Womersley's symmetric designs ($N \approx t^2/2$) for every odd $t \leq 51$, and 26 of his 30 non-symmetric designs ($N \approx t^2/2$ as well, for $t \leq 30$). We are not aware of earlier rigorous existence proofs in this regime. The pipeline regenerates every verdict in under a minute on a laptop; we close with recommendations for evaluators and for reporting records.

1 Introduction

Since FunSearch and AlphaEvolve (Romera-Paredes et al., 2024; Novikov et al., 2025; Georgiev et al., 2025), LLM-guided program search has produced a stream of improved bounds: autoconvolution constants, the Erdős minimum overlap problem, packings of circles, hexagons and cubes, kissing configurations, spherical designs and matrix-multiplication tensor decompositions. Open-source reimplementations and new methods followed (Sharma, 2025; Lange et al., 2025; Wang et al., 2025; Yuksekgonul et al., 2026; Ye et al., 2026), and since March 2026 a public leaderboard lets AI agents compete on the same problems (Bianchi et al., 2026). Improvements are often tiny, sometimes below 10^{-8}, and are reported as new records.

Every one of these systems scores candidates with a floating-point program, and several evaluators add explicit tolerances: 10^{-6} on overlaps, 10^{-4} on self-reported scores, 10^{-3} on constraints. The problems themselves are exact: a packing either overlaps or it does not, and a bound either holds or it does not. Critiques of this literature have focused on search budgets and baselines (Gideoni et al., 2026; Berthold et al., 2026b; Gupta et al., 2026) or on individual evaluator issues (Wang et al., 2025; Bianchi et al., 2026; Chung et al., 2026). To our knowledge no one has taken the *published artifacts* and asked, uniformly and exactly: is each construction valid, does it establish the printed value, and does it beat the best construction published before it?

Contributions.

1. An exact audit of 115 claims from 10 sources, under two exact readings of every published number (the printed decimal, and the binary64 value the authors' code used), with verdicts VALID, REPAIRABLE (with a certified repair) or INVALID (§3, §4.1).
2. A claim-level analysis: unsafe roundings, records that do not beat the prior public construction, and definitional mismatches (§4.2).
3. An evaluator analysis: tolerances across systems, and explicit inputs that published evaluators accept with inflated scores (§4.3).
4. Certified mathematics: certified values for all audited constructions, and Krawczyk existence proofs for 61 efficient spherical designs (§4.4).
5. A reproducible pipeline and recommendations (§5).

2 Background

Evaluators. AlphaEvolve's problem repository (Google DeepMind, 2025) documents how evolved programs exploit naive score functions (negative values, underflow, complex inputs), and evaluators were hardened in response. ThetaEvolve (Wang et al., 2025) tabulates circle-packing tolerances and notes that AlphaEvolve's third-autocorrelation code uses $|\max f*f|$ while the problem is stated with $\max |f*f|$. EinsteinArena (Bianchi et al., 2026) moved its kissing verifier to 80-digit arithmetic and later removed slack from packing verifiers after an internal audit. SimpleTES (Ye et al., 2026) documents a tolerance-hacked circle packing. Station (Chung et al., 2026) separates sampled scores from proven inequalities; Berthold et al. (Berthold et al., 2026a) post-process solver output in arbitrary precision.

Rigorous numerics. We use Arb ball arithmetic (Johansson, 2016). Our design certificates follow the interval-Newton approach (Krawczyk, 1969) used by Chen, Frommer and Lang (Chen et al., 2011) for designs with $N = (t+1)^2$ points. Efficient designs with $N \approx t^2/2$ (Hardin & Sloane, 1996; Womersley, 2018; 2016) have so far been numerical only.

3 Method

Claims and artifacts. For each claim we record the problem, the printed value and its direction (upper or lower bound, maximisation or minimisation), the prior record it claims to beat, and the artifact's provenance (repository, commit, file and key). Numbers are extracted as *source text*: notebooks are parsed with an AST and JSON with numbers kept as strings, so nothing is rounded before verification.

Two exact readings. Each number is read as the exact rational value of its decimal text and, separately, as the exact value of `float(text)`. The two differ by at most one ulp but can lie on different sides of a constraint.

Exact evaluation. Verdicts are computed only in exact rationals or integers, or as Arb enclosures. Polynomial feasibility is checked in squared form, and irrational distances are bracketed with integer square roots. For an equal-step function f with heights $a_0, \dots, a_{n-1}$, $f*f$ is piecewise linear with knot values proportional to the discrete self-convolution $b = a * a$. Hence

$$C_1 \le \frac{2n \max_k b_k}{(\sum_i a_i)^2}, \qquad C_2 \ge \frac{\sum_k (y_k^2 + y_k y_{k+1} + y_{k+1}^2)}{3\,(\sum_k b_k)(\max_k b_k)}, \quad y = (0, b, 0),$$

and $C_3 \le 2n\,|\max_k b_k|/(\sum a)^2$, all evaluated exactly with FLINT polynomial multiplication. For the Erdős overlap the maximum over continuous shifts is attained at grid shifts. Planar constraints involving $\sqrt{3}$ are decided by squaring. Rotated polytopes are certified by separating axes, using *exact rational* axis vectors applied to interval vertices. Kissing configurations with coordinates in $\mathbb{Z}[\sqrt{2}]$ are checked in $\mathbb{Z}[\sqrt{2}]$, and tensor decompositions are reconstructed over the Gaussian rationals.

Verdicts and certified repairs. VALID means every constraint holds exactly. REPAIRABLE means some constraint fails and an explicit modification is *proven* to restore feasibility: shrinking all radii by a proven bound on the worst violation, rescaling to the exact perimeter, redistributing mass within $[0, 1]$, or shrinking towards a centroid with a rational bound on the area loss. The certified score is then a rigorous bound on what the published construction achieves. Anything else is INVALID.

Claim checks. "Claim holds" compares the exact (or certified) value with the printed one in the safe direction. A failure is FLOAT_NOISE if $|\text{exact} - \text{printed}| \leq 10^{-12}$ relative and MISSTATED otherwise. "Beats prior" compares with the best previously *released* construction, dated by repository history and arXiv versions, not by the rounded headline the claimant cites.

Existence proofs for spherical designs. An antipodal set on S^2 is a t-design (t odd) iff its degree-$(t-1)$ monomial moments equal the sphere moments $\mathbb{E}[x^{2a}y^{2b}z^{2c}] = \frac{(2a-1)!!(2b-1)!!(2c-1)!!}{(2(a+b+c)+1)!!}$. We proceed as follows.

1. Refine the published points to 90 digits by Gauss–Newton.
2. Take the rank $r = \min(\#\text{eq} - 1, \#\text{unknowns} - 3)$ (one exact dependency, three rotation gauges), unless the float singular spectrum shows a clear larger deficit.
3. Select r equations and r free angles by pivoted QR, and fix the remaining angles at exact rationals.
4. Run Newton steps with an Arb Jacobian, then the Krawczyk test (Krawczyk, 1969) in 400-bit Arb on a box of radius 10^{-40}.

Lemma 1. *If the selected r equations have a unique zero in the box (Krawczyk inclusion) and exactly one equation was dropped, the zero is an exact spherical t-design.*

Proof sketch. On S^2, $(x^2 + y^2 + z^2)^{(t-1)/2} \equiv 1$ gives a fixed linear relation among the moment equations. Since the selected rows have an invertible Jacobian at the zero, the relation must involve the dropped row, which therefore vanishes there too. □

For general (non-antipodal) sets we use the monomials of degrees t and $t-1$. Together they represent every spherical harmonic of degree $\leq t$ exactly once, $(t+1)^2$ equations in $2N$ angles, and the lemma holds with the same sphere identity applied to whichever of the two degrees is even.

Configurations whose Jacobian is rank-deficient because of a symmetry $C_k \times \{\pm I\}$ (antipodal) or C_k (general) are certified in the invariant subspace. The unknowns are the angles of one point per orbit, and the equations are the C_k-invariant harmonic moments, whose number n_{inv} we count exactly. For $t = 15$, $k = 3$ antipodal there are 39 invariant harmonics and 39 free angles. The system is square, and its zeros are designs because non-invariant moments vanish by group averaging. The monomial residuals equal $L\,h$ for a fixed linear map L and the vector h of n_{inv} invariant moments. If exactly n_{inv} rows pass the Krawczyk test, the selected block of L is invertible, so $h = 0$. The choices of subsystem and preconditioner are made in floating point, but they affect only *whether* a proof succeeds, never its correctness.

4 Results

4.1 RQ1: are the constructions valid?

Table 1 gives per-source tallies. All 15 AlphaEvolve tensor decompositions are exactly correct over their stated rings, as are the 593-point kissing configuration and every integer construction (Kakeya sets, sum-difference sets, isosceles-free and 5-cospherical-free point sets). The four REPAIRABLE AlphaEvolve cases all come from zero-tolerance floating-point checks on irrational or rounded data:

- circle packings with overlaps of about 5×10^{-17};
- a Heilbronn point 2×10^{-17} outside an edge of irrational slope;

Source	Claims	Valid	Repair.	Invalid	Holds	Float noise	Misstated	Not a record
AlphaEvolve repository	25	23	2	0	19	0	0	0
AlphaEvolve results notebook	31	29	2	0	31	0	0	0
Berthold / FICO Xpress	4	4	0	0	4	0	0	0
Boyer-Li	1	1	0	0	1	0	0	0
Discovery Loop	12	12	0	0	0	0	0	0
EinsteinArena leaderboard	18	11	6	1	6	10	2	2
ShinkaEvolve	1	1	0	0	1	0	0	0
SimpleTES	7	4	3	0	2	0	1	2
TTT-Discover	4	3	1	0	3	0	1	0
ThetaEvolve	3	2	1	0	1	0	2	2
Together AI	9	7	2	0	7	0	2	1
Total	115	97	17	1	75	10	8	7

Table 1: Verdicts by source (worst status over the two readings). "Holds" means the printed value is established exactly. "Float noise" means it fails only in the last digits of a float printout. "Not a record" means the certified value does not exceed the prior public construction.

- an Erdős construction whose heights sum to $n/2 - 2.4 \times 10^{-15}$, where the notebook check `np.sum(h) == n/2` passes only by rounding.

The earlier 2025 packings are exactly valid; the later, higher-scoring versions are not.

4.2 RQ2: do the claims hold?

Table 2 lists printed claims that the released constructions do not establish. TTT-Discover states twice in its text that its 30,000-step function "certifies that the upper bound $C_1 \leq 1.50286$"; the function gives exactly $C_1 = 1.5028628982\ldots$. Its table value of 1.50287 is correct. Similar unsafe roundings affect SimpleTES's C_3, ThetaEvolve's 26-circle packing and Together AI's Tammes value.

Table 3 lists claimed improvements that do not exceed an earlier public construction. ThetaEvolve's 26-circle "new best-known bound" (28 Nov 2025) is 7.5×10^{-9} below a construction that has been in AlphaEvolve's repository since its initial commit on 5 Nov 2025; that construction certifies at 2.63598308491760 even after its own 10^{-17} repair. AlphaEvolve's $t = 19$ design with $N = 198$ is not an improvement on Womersley's $N = 192$ (Womersley, 2016), and its record for sum-difference problem I was below existing literature (Penman & Wells, 2013); the constant has since been shown to equal 2 (Lin & Li, 2026).

Definitional mismatch. AlphaEvolve's construction behind "$C_3 \leq 1.4557$" gives 1.4556428 for $|\max f*f|$, the definition in its code, but 4.334 for $\max |f*f|$, the definition in its problem registry. The registry's record is instead supported by a different construction ($1.4687621 < 1.4993$).

Leaderboard artifacts. The public Tammes verifier clamps norms below 10^{-12} before normalising. Its current leader (displayed 0.5633) submits vectors of norm $\approx 10^{-13}$ that are never projected onto the sphere; projected correctly, the minimum distance is 0.2140. The runner-up contains the zero vector. The genuine leader is third, at 0.51347208468056467 exactly.

4.3 RQ3: evaluators

Table 4 collects evaluator tolerances quoted from source. Table 5 shows inputs that OpenEvolve's own verification functions accept.

- Inflating every radius of AlphaEvolve's 26-circle packing by 4.99×10^{-7} is accepted at 2.6359960589, which is 1.3×10^{-5} above the true best and reproduces the value SimpleTES reports as a tolerance hack. Exact verification certifies it back to 2.6359830849.

Claim	Direction	Printed	Exact / certified	Exact−printed
TTT-AC1-text	upper	1.50286	1.5028628982558265	2.90×10^{-6}
THETA-CP26-formal	max	2.63598308	2.6359830773881193	-2.61×10^{-9}
THETA-CP26-tol	max	2.6359857	2.6359816383240308	-4.06×10^{-6}
TA-TAMMES	max	0.5134721	0.513472084680564	-1.53×10^{-8}
TA-CR21	max	2.3658323759	2.3658323758394027	-6.06×10^{-11}
EA-TAMMES-2497	max	0.5633081876528571	0.213968797876115	-3.49×10^{-1}
EA-TAMMES-2496	max	0.516349728124321	invalid (zero vector)	—
ST-AC3	upper	1.453675	1.4536754655951828	4.66×10^{-7}

Table 2: Printed claims not established by the released construction.

Claim	Certified value	Prior public construction	Source of prior
THETA-CP26-formal	2.6359830773881193	2.6359830849176055	AE repository construction
THETA-CP26-tol	2.6359816383240308	2.6359830849176055	AE repository construction
TA-CR21	2.3658323758394027	2.3658323759108107	AE repository construction
EA-TAMMES-2497	0.213968797876115	0.5134720846805647	KawaiiCorgi
EA-CR21-2544	2.3658323759098584	2.3658323759108107	AE repository construction
ST-CP26	2.6359830849080790	2.6359830849176055	AE repository
ST-CP32	2.9395727711955525	2.9395727712063213	AE repository

Table 3: Claims whose certified value does not exceed the prior public construction.

- Reporting the unchanged best Erdős construction as its computed value minus 0.99×10^{-4} is accepted at 0.3807596. That "beats" the genuine record by 9.9×10^{-5}, more than all progress since 2016 (6.8×10^{-5}, (Haugland, 2016)).
- Naively exploiting the 10^{-3} mass slack by scaling h does not help (a negative result).

Figure 1 compares the margins of verified improvements with these tolerances: 5 of 49 are below 10^{-6}.

4.4 Certified spherical designs

All 12 AlphaEvolve designs ($t = 15, 19, 21$) and all 23 Womersley symmetric designs for odd t from 7 to 51 (up to $N = 1328$) are proven to exist within 10^{-40} of the published points (Table 6, Appendix). Krawczyk ratios range from 10^{-37} to 10^{-27}, and proof times from 0.1 s ($t = 9$) to 18 min ($t = 51$) on a laptop. Both $(15, 120)$ configurations have a rank-deficient Jacobian caused by a $C_3 \times \{\pm I\}$ symmetry and were proven in the invariant subspace.

Womersley's non-symmetric designs (Womersley, 2018) ($t \leq 30$, Table 7) show two regimes.

- For $t = 1$, every even $t \geq 6$ and every $t \geq 16$, $2N - 3 \geq (t + 1)^2 - 1$, and the square method proves every design (N up to 482; ratios 10^{-40} to 10^{-30}).
- For odd $t \leq 15$ the point count is smaller than the naive count, $2N-3 = (t+1)^2-3$. The full system is *overdetermined*, so no square subsystem can certify it. Their existence must come from structure, here symmetry. Each is invariant under a 2-fold rotation, and in the C_2-invariant subspace the system is exactly square ($N-1$ angles against $N-1$ invariant harmonics). All four ($t = 9, 11, 13, 15$) are proven this way, as is the octahedron ($t = 2$) under C_3.

The four exceptions, $t = 3, 4, 5, 7$ ($N \leq 32$), need larger symmetry groups or have points on the rotation axis, so our orbit parametrisation does not apply to them. In total, 26 of the 30 non-symmetric designs are proven.

Two numerical lessons emerged at larger t.

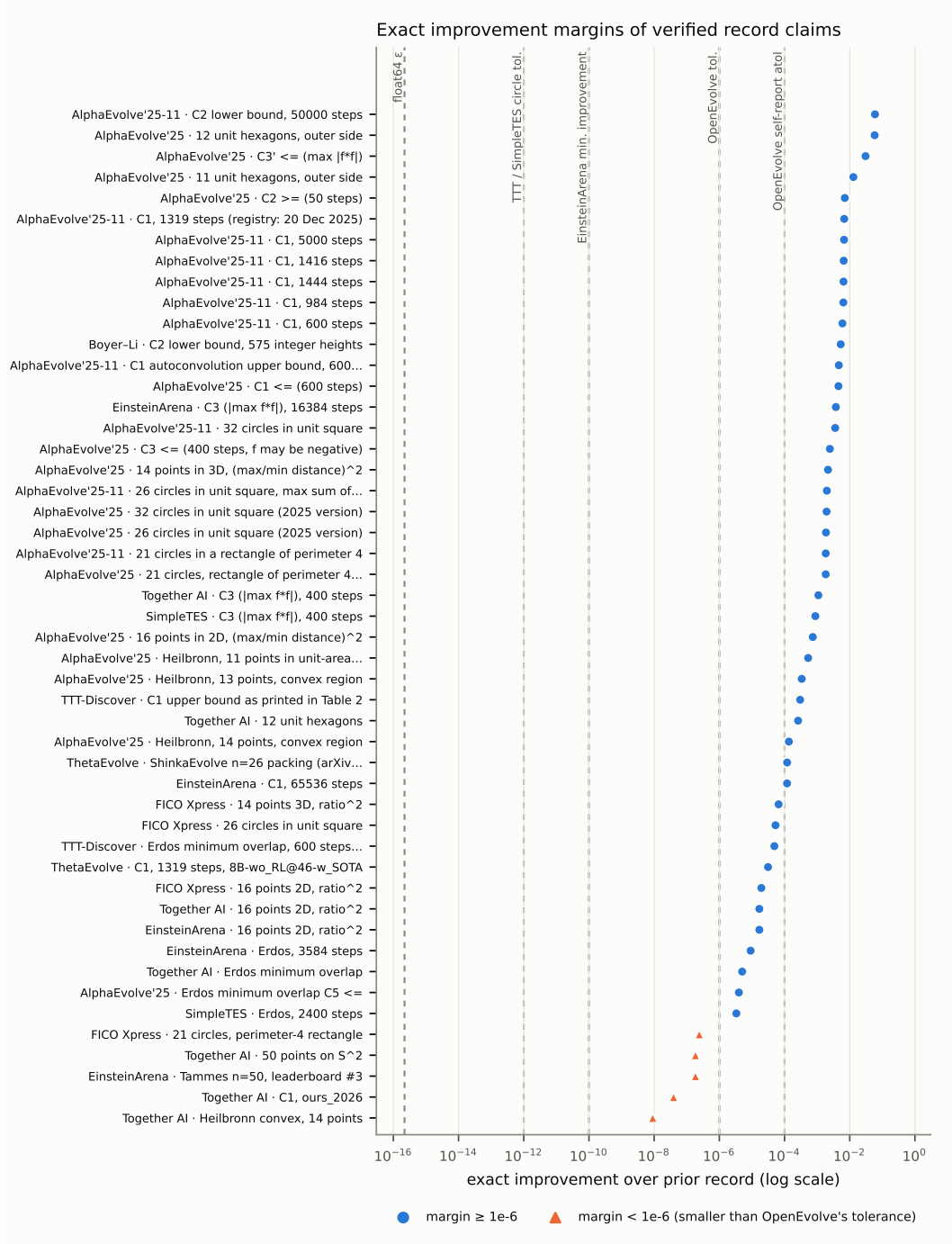


Figure 1: Exact improvement margins of verified record claims (kissing counts excluded), with float64 ε and published evaluator tolerances.

Evaluator	Problem	Tolerance / check
OpenEvolve (and ThetaEvolve)	circle packing	overlap and containment slack 10^{-6}
ShinkaEvolve (paper run)	circle packing	OpenEvolve verifier, slack 10^{-6}; current code `atol`= 0 but `np.isclose` rtol 10^{-5} on sums
TTT-Discover, SimpleTES	circle packing	slack 10^{-12}
OpenEvolve	Erdős overlap	$\int h$ within 10^{-3}, no renormalisation; self-reported score within 10^{-4}
TTT-Discover	Erdős overlap	self-reported score within 10^{-4} (renormalises h)
SimpleTES	third autocorrelation	self-reported C_3 within 10^{-3}
AlphaEvolve (2025 notebook)	Erdős overlap	float equality `np.sum(h) == n/2`
AlphaEvolve (repository)	no-5-on-a-sphere	float 5×5 determinant, `np.isclose(det, 0)`
AlphaEvolve (repository)	rectangle packing	perimeter via `np.isclose` (rtol 10^{-5})
EinsteinArena	Tammes	norms below 10^{-12} clamped to 10^{-12} before normalising
EinsteinArena	all	minimum improvement 10^{-10} to 10^{-5}; slack removed from packings after internal audit (Aug 2026)

Table 4: Tolerances and checks in published evaluators.

Evaluator	Modification	Accepted score	Exact value
OpenEvolve circle packing	every radius += 4.99e-07	2.6359960589	2.635983084917605
OpenEvolve Erdős overlap	report c5 = computed - 0.99e-4 (the construction is unchanged)	0.3807595749	$3.808585748578585 \times 10^{-1}$

Table 5: Inputs accepted by published evaluators with inflated scores, and their exact values.

- Genuine singular values decay to about 10^{-12} relative by $t = 27$, so fixed rank thresholds misclassify them.
- From $t = 41$ they fall below float64 resolution, so the refinement must use the *a-priori* rank and finish with Newton steps whose Jacobian is solved in Arb.

5 Recommendations

1. Certify the final claim with exact or interval evaluation; floating point is fine inside the search.
2. Always recompute scores; never score self-reported values.
3. Round printed bounds in the safe direction, with enough digits to separate the claim from the prior record.
4. Release exact artifacts, integers or rationals where possible (as Boyer–Li (Boyer & Li, 2025) and Discovery Loop (Sander, 2026) do), with the claimed value computed from them.
5. Compare against the best *released* construction, dated by public release, not against a rounded headline.
6. Leaderboards: reject degenerate inputs, check feasibility exactly with zero slack, and require margins above the verifier's resolution.

6 Limitations

We audit public artifacts only; several claims have no released coordinates. A "printed value" depends on the sentence quoted, so we archive each quote. Priority dating relies on git and arXiv timestamps. The

design proofs certify existence near the published points, not optimality of N. The Penman–Wells value is taken from a public issue report rather than recomputed. We did not attempt matched-budget re-runs of the search methods.

Reproducibility

All code, registries, source snapshots (with URLs and fetch dates), results and logs are in the accompanying artifact. `scripts/run_all.py` verifies all 115 claims in about 40 s. The design proofs take about 1 min (AlphaEvolve), 56 min (Womersley symmetric) and 36 min (Womersley non-symmetric). Tables in this paper are generated by `scripts/make_tables.py`.

References

Timo Berthold, Dominik Kamp, Gioni Mexi, Sebastian Pokutta, and Imre Polik. Out-of-the-Box Global Optimization for Packing Problems: New Models and Improved Solutions. *arXiv preprint arXiv:2605.04850*, 2026a.

Timo Berthold, Dominik Kamp, Gioni Mexi, Sebastian Pokutta, and Imre Pólik. Global Optimization for Combinatorial Geometry Problems Revisited in the Era of LLMs. *arXiv preprint arXiv:2601.05943*, 2026b.

Federico Bianchi, Yongchan Kwon, Aneesh Pappu, and James Zou. Harnessing the Collective Intelligence of AI Agents in the Wild for New Discoveries. *arXiv preprint arXiv:2606.10402*, 2026.

Christopher Boyer and Zane Kun Li. An improved example for an autoconvolution inequality. *arXiv preprint arXiv:2506.16750*, 2025.

Xiaojun Chen, Andreas Frommer, and Bruno Lang. Computational existence proofs for spherical t-designs. *Numerische Mathematik*, 117:289–305, 2011. doi: 10.1007/s00211-010-0332-5.

Stephen Chung, Wenyu Du, and William J. Wesley. Autonomous Mathematical Discovery in an Open-World Multi-Agent Environment. *arXiv preprint arXiv:2608.23691*, 2026.

Bogdan Georgiev, Javier Gómez-Serrano, Terence Tao, and Adam Zsolt Wagner. Mathematical exploration and discovery at scale. *arXiv preprint arXiv:2511.02864*, 2025.

Yonatan Gideoni, Sebastian Risi, and Yarin Gal. Simple Baselines are Competitive with Code Evolution. *arXiv preprint arXiv:2602.16805*, 2026.

Google DeepMind. Mathematical problem repository for AlphaEvolve. `https://github.com/google-deepmind/alphaevolve_repository_of_problems`, 2025. Initial commit 5 Nov 2025; commit 8f44745 accessed 23 Sep 2026.

Akshat Gupta, Jermaine Lei, Alexander Lu, Gopala Anumanchipalli, and Leshem Choshen. Automated Discovery Has No Universally Superior Harness. *arXiv preprint arXiv:2607.18235*, 2026.

R. H. Hardin and N. J. A. Sloane. Mclaren's improved snub cube and other new spherical designs in three dimensions. *Discrete & Computational Geometry*, 15:429–441, 1996. doi: 10.1007/BF02711518.

Jan Kristian Haugland. The minimum overlap problem revisited. *arXiv preprint arXiv:1609.08000*, 2016.

Fredrik Johansson. Arb: Efficient Arbitrary-Precision Midpoint-Radius Interval Arithmetic. *arXiv preprint arXiv:1611.02831*, 2016.

Rudolf Krawczyk. Newton-algorithmen zur bestimmung von nullstellen mit fehlerschranken. *Computing*, 4: 187–201, 1969. doi: 10.1007/BF02234767.

Robert Tjarko Lange, Yuki Imajuku, and Edoardo Cetin. ShinkaEvolve: Towards Open-Ended And Sample-Efficient Program Evolution. *arXiv preprint arXiv:2509.19349*, 2025.

Haowei Lin and Shanda Li. Settling the Optimal Exponent Relating Sumsets and Difference Sets. *arXiv preprint arXiv:2607.27199*, 2026.

Alexander Novikov, Ngân V, Marvin Eisenberger, Emilien Dupont, Po-Sen Huang, Adam Zsolt Wagner, Sergey Shirobokov, Borislav Kozlovskii, Francisco J. R. Ruiz, Abbas Mehrabian, M. Pawan Kumar, Abigail See, Swarat Chaudhuri, George Holland, Alex Davies, Sebastian Nowozin, Pushmeet Kohli, and Matej Balog. AlphaEvolve: A coding agent for scientific and algorithmic discovery. *arXiv preprint arXiv:2506.13131*, 2025.

David Penman and Matthew Wells. On sets with more restricted sums than differences. *Integers*, 13:A57, 2013.

Bernardino Romera-Paredes, Mohammadamin Barekatain, Alexander Novikov, Matej Balog, M. Pawan Kumar, Emilien Dupont, Francisco J. R. Ruiz, Jordan S. Ellenberg, Pengming Wang, Omar Fawzi, Pushmeet Kohli, and Alhussein Fawzi. Mathematical discoveries from program search with large language models. *Nature*, 625:468–475, 2024. doi: 10.1038/s41586-023-06924-6.

Wes Sander. LLM-Guided Program Evolution for Circle Packing: Breaking 10 Packomania Records for $28. *arXiv preprint arXiv:2609.05093*, 2026.

Asankhaya Sharma. OpenEvolve: an open-source implementation of AlphaEvolve. `https://github.com/algorithmicsuperintelligence/openevolve`, 2025. Accessed 24 Sep 2026.

Yiping Wang, Shao-Rong Su, Zhiyuan Zeng, Eva Xu, Liliang Ren, Xinyu Yang, Zeyi Huang, Xuehai He, Luyao Ma, Baolin Peng, Hao Cheng, Pengcheng He, Weizhu Chen, Shuohang Wang, Simon Shaolei Du, and Yelong Shen. ThetaEvolve: Test-time Learning on Open Problems. *arXiv preprint arXiv:2511.23473*, 2025.

Robert S. Womersley. Symmetric spherical designs on the sphere S^2 with good geometric properties. `https://web.maths.unsw.edu.au/~rsw/Sphere/EffSphDes/ss.html`, 2016. Last updated 27 Sep 2016; accessed 24 Sep 2026.

Robert S. Womersley. Efficient spherical designs with good geometric properties. In *Contemporary Computational Mathematics – A Celebration of the 80th Birthday of Ian Sloan*, pp. 1243–1285. Springer, 2018. doi: 10.1007/978-3-319-72456-0_57.

Haotian Ye, Haowei Lin, Jingyi Tang, Yizhen Luo, Rahul Thapa, Caiyin Yang, Chang Su, Rui Yang, Ruihua Liu, Rundao Li, Zeyu Li, Pengwei Sun, Chong Gao, Dachao Ding, Guangrong He, Miaolei Zhang, Lina Sun, Wenyang Wang, Yuchen Zhong, Zhuohao Shen, Puheng Li, Pan Lu, Bianxiao Cui, Di He, Jianzhu Ma, Junfeng Li, Hexi Baoyin, Yejin Choi, Stefano Ermon, Xiaowen Chu, Tongyang Li, Yuzhi Xu, and James Zou. Structured Scaling of AI Discovery Across Diverse Scientific Domains. *arXiv preprint arXiv:2604.19341*, 2026.

Mert Yuksekgonul, Daniel Koceja, Xinhao Li, Federico Bianchi, Jed McCaleb, Xiaolong Wang, Jan Kautz, Yejin Choi, James Zou, Carlos Guestrin, and Yu Sun. Learning to Discover at Test Time. *arXiv preprint arXiv:2601.16175*, 2026.

A Design certificates

t	N	Source	rank	method	$\max \lvert K - \tilde{x} \rvert / r$	time (s)
7	32	Womersley	27	square	2.1×10^{-37}	0.3
9	48	Womersley	44	square	1.3×10^{-36}	0.1
11	70	Womersley	65	square	8.3×10^{-36}	0.1
13	94	Womersley	90	square	5.2×10^{-35}	0.2
15	120	AlphaEvolve	39	sym. C_3	3.9×10^{-35}	51.8
15	120	Womersley	39	sym. C_3	3.1×10^{-35}	24.9
15	124	AlphaEvolve	119	square	8.1×10^{-35}	0.5
15	126	AlphaEvolve	119	square	5.7×10^{-35}	0.6
15	128	AlphaEvolve	119	square	3.0×10^{-35}	0.6
15	130	AlphaEvolve	119	square	2.6×10^{-35}	0.5
17	156	Womersley	152	square	2.0×10^{-34}	0.9
19	192	Womersley	189	square	1.2×10^{-33}	1.5
19	198	AlphaEvolve	189	square	3.3×10^{-34}	1.5
19	200	AlphaEvolve	189	square	3.3×10^{-34}	1.5
19	202	AlphaEvolve	189	square	2.8×10^{-34}	1.5
19	204	AlphaEvolve	189	square	1.4×10^{-34}	1.7
21	234	AlphaEvolve	230	square	1.7×10^{-33}	2.5
21	234	Womersley	230	square	3.9×10^{-33}	2.4
21	236	AlphaEvolve	230	square	1.2×10^{-33}	2.5
21	238	AlphaEvolve	230	square	6.1×10^{-34}	2.5
23	278	Womersley	275	square	5.0×10^{-33}	4.0
25	328	Womersley	324	square	1.2×10^{-32}	5.9
27	380	Womersley	377	square	5.6×10^{-32}	9.2
29	438	Womersley	434	square	4.5×10^{-31}	12.1
31	498	Womersley	495	square	2.3×10^{-31}	17.6
33	564	Womersley	560	square	1.1×10^{-30}	24.1
35	632	Womersley	629	square	2.1×10^{-30}	34.1
37	706	Womersley	702	square	2.5×10^{-30}	47.7
39	782	Womersley	779	square	6.0×10^{-30}	66.2
41	864	Womersley	860	square	3.7×10^{-28}	226.4
43	948	Womersley	945	square	8.7×10^{-29}	284.9
45	1038	Womersley	1034	square	1.0×10^{-28}	360.1
47	1130	Womersley	1127	square	5.8×10^{-28}	446.4
49	1228	Womersley	1224	square	3.2×10^{-28}	702.2
51	1328	Womersley	1325	square	3.9×10^{-27}	1099.8

Table 6: Krawczyk existence certificates for spherical t-designs (box radius 10^{-40}, 400-bit Arb).

t	N	$2N-3$	$(t+1)^2-1$	status	method	$\max\lvert K-\tilde{x}\rvert/r$	time (s)
1	3	3	3	proven	square	3.2×10^{-40}	0.7
2	6	9	8	proven	sym. C_3	2.9×10^{-40}	0.4
3	8	13	15	not proven	square	—	0.3
4	14	25	24	not proven	square	—	0.0
5	18	33	35	not proven	square	—	2.8
6	26	49	48	proven	square	1.9×10^{-36}	0.1
7	32	61	63	not proven	square	—	9.0
8	42	81	80	proven	square	3.7×10^{-35}	0.4
9	50	97	99	proven	sym. C_2	8.9×10^{-36}	37.4
10	62	121	120	proven	square	9.8×10^{-35}	1.1
11	72	141	143	proven	sym. C_2	3.6×10^{-35}	113.8
12	86	169	168	proven	square	2.6×10^{-34}	2.4
13	98	193	195	proven	sym. C_2	5.9×10^{-35}	291.6
14	114	225	224	proven	square	4.4×10^{-34}	4.7
15	128	253	255	proven	sym. C_2	1.1×10^{-34}	660.7
16	146	289	288	proven	square	2.9×10^{-33}	8.7
17	163	323	323	proven	square	4.6×10^{-33}	11.2
18	182	361	360	proven	square	5.5×10^{-33}	14.6
19	201	399	399	proven	square	4.6×10^{-33}	20.2
20	222	441	440	proven	square	1.1×10^{-32}	25.3
21	243	483	483	proven	square	1.3×10^{-32}	32.0
22	266	529	528	proven	square	2.4×10^{-32}	40.1
23	289	575	575	proven	square	1.5×10^{-32}	49.4
24	314	625	624	proven	square	1.0×10^{-31}	62.3
25	339	675	675	proven	square	1.5×10^{-31}	74.1
26	366	729	728	proven	square	6.3×10^{-31}	90.8
27	393	783	783	proven	square	5.0×10^{-31}	112.3
28	422	841	840	proven	square	4.2×10^{-31}	137.4
29	451	899	899	proven	square	1.6×10^{-30}	167.2
30	482	961	960	proven	square	1.4×10^{-30}	198.6

Table 7: Womersley's non-symmetric designs (Womersley, 2018). $2N-3$ free angles against $(t+1)^2-1$ independent moment conditions. Where the second exceeds the first, the system is overdetermined, and a proof needs the symmetric reduction. "Not proven" means the square subsystem has a certified zero but the remaining equations are not certified.